

Order Management System

Week -4

1. Database Design

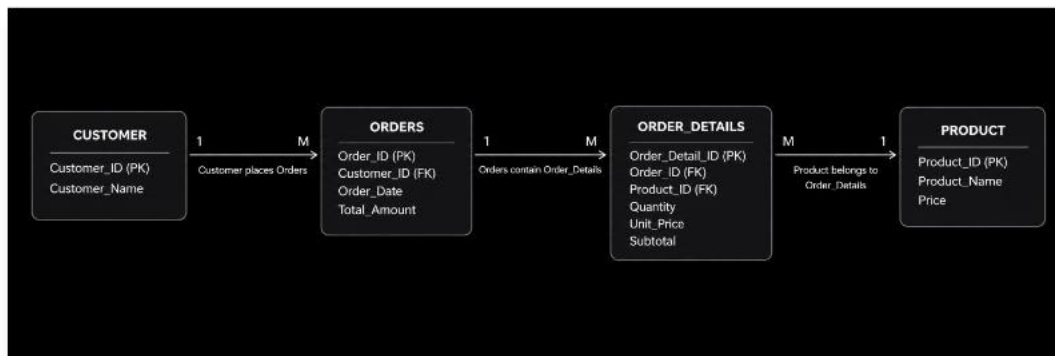
1.1 Orders Table

Attribute	Data Type	Constraint
Order_ID	INT	Primary Key
Customer_ID	INT	Foreign Key
Order_Date	DATE	NOT NULL
Total_Amount	NUMBER(10,2)	NOT NULL

1.2 Order_Details Table

Attribute	Data Type	Constraint
Order_Detail_ID	INT	Primary Key
Order_ID	INT	Foreign Key
Product_ID	INT	Foreign Key
Quantity	INT	NOT NULL
Unit_Price	NUMBER(10,2)	NOT NULL
Subtotal	NUMBER(10,2)	NOT NULL

2. Table Relationships



3. Table Creation

3.1 Create Orders Table

```
SQL> CREATE TABLE Orders (  
2     Order_ID INT PRIMARY KEY,  
3     Customer_ID INT,  
4     Order_Date DATE NOT NULL,  
5     Total_Amount NUMBER(10,2) NOT NULL,  
6     FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)  
7 );
```

Table created.

3.2 Create Order_Details

```
SQL> CREATE TABLE Order_Details (  
 2     Order_Detail_ID INT PRIMARY KEY,  
 3     Order_ID INT,  
 4     Product_ID INT,  
 5     Quantity INT NOT NULL,  
 6     Unit_Price NUMBER(10,2) NOT NULL,  
 7     Subtotal NUMBER(10,2) NOT NULL,  
 8     FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID),  
 9     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)  
10 );
```

Table created.

4. Order Data Insertion

```
SQL> INSERT INTO Orders  
 2 (Order_ID, Customer_ID, Order_Date, Total_Amount)  
 3 VALUES  
 4 (501, 201, SYSDATE, 79999.00);
```

1 row created.

```
SQL>  
SQL> INSERT INTO Orders  
 2 (Order_ID, Customer_ID, Order_Date, Total_Amount)  
 3 VALUES  
 4 (502, 202, SYSDATE, 799.00);
```

1 row created.

```
SQL>  
SQL> INSERT INTO Orders  
 2 (Order_ID, Customer_ID, Order_Date, Total_Amount)  
 3 VALUES  
 4 (503, 203, SYSDATE, 1999.00);
```

1 row created.

5. Order_Details Data Insertion

```

SQL> INSERT INTO Order_Details
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3 VALUES
  4 (1003, 503, 104, 1, 45999.00, 45999.00);

1 row created.

SQL> INSERT INTO Order_Details
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3 VALUES
  4 (1004, 501, 102, 1, 64999.00, 64999.00);

1 row created.

SQL> INSERT INTO Order_Details
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3 VALUES
  4 (1005, 502, 103, 1, 24999.00, 24999.00);

1 row created.

```

6. Order Modification Operations

6.1 Modify Order Amount

```

SQL> UPDATE Orders
  2 SET Total_Amount = 84999.00
  3 WHERE Order_ID = 501;

1 row updated.

```

```

SQL> SELECT * FROM Orders
  2 WHERE Order_ID = 501;

ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
-----
      501          201 24-SEP-26          84999

```

6.2 Modify Order Date

```
SQL> UPDATE Orders
  2  SET Order_Date = SYSDATE
  3  WHERE Order_ID = 502;

1 row updated.
```

7. Order_Details Modification

```
SQL> UPDATE Order_Details
  2  SET Quantity = 2,
  3      Subtotal = 91998.00
  4  WHERE Order_Detail_ID = 1003;

1 row updated.
```

8. Customer Order History Report

```
SQL> SELECT
  2     C.Customer_ID,
  3     C.First_Name,
  4     C.Last_Name,
  5     O.Order_ID,
  6     O.Order_Date,
  7     P.Product_Name,
  8     OD.Quantity,
  9     OD.Unit_Price,
 10     OD.Subtotal,
 11     O.Total_Amount
 12 FROM Customer C
 13 JOIN Orders O
 14     ON C.Customer_ID = O.Customer_ID
 15 JOIN Order_Details OD
 16     ON O.Order_ID = OD.Order_ID
 17 JOIN Product P
 18     ON OD.Product_ID = P.Product_ID
 19 ORDER BY C.Customer_ID, O.Order_Date;
```

CUSTOMER_ID FIRST_NAME

LAST_NAME ORDER_ID ORDER_DAT

PRODUCT_NAME

QUANTITY UNIT_PRICE SUBTOTAL TOTAL_AMOUNT

```

      201 Arun
Kumar
iPhone 15
      1      79999      79999      84999
      501 24-SEP-26
```

CUSTOMER_ID FIRST_NAME

9. Customer-wise Order Summary

```
SQL> SELECT
  2     C.Customer_ID,
  3     C.First_Name,
  4     C.Last_Name,
  5     COUNT(O.Order_ID) AS Total_Orders,
  6     SUM(O.Total_Amount) AS Total_Amount
  7 FROM Customer C
  8 LEFT JOIN Orders O
  9     ON C.Customer_ID = O.Customer_ID
 10 GROUP BY
 11     C.Customer_ID,
 12     C.First_Name,
 13     C.Last_Name
 14 ORDER BY C.Customer_ID;
```

CUSTOMER_ID FIRST_NAME

LAST_NAME TOTAL_ORDERS TOTAL_AMOUNT

```

      201 Arun
Kumar
      1      84999
      202 Priya
Sharma
      1        799
      203 Rahul
Raj
      1      1999
```

CUSTOMER_ID FIRST_NAME

LAST_NAME TOTAL_ORDERS TOTAL_AMOUNT

```

      204 Keerthi
Kumar
      0
```